StrawMind: An Attention-Augmented Real-Time Detector and a Reproducible Imbalanced Benchmark for Mushroom Fungal-Disease Detection

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Abstract

Early detection of fungal contamination (*Trichoderma* green mold and *Aspergillus*) is critical for mushroom food safety and yield, yet existing computer-vision studies lack a reproducible, class-balanced benchmark and rarely address the severe class imbalance that characterizes real cultivation imagery. We present StrawMind, comprising (i) a curated three-class detection benchmark (Healthy, *Trichoderma*, *Aspergillus*) built from two public sources with a strict stratified 70/15/15 split and an absolutely held-out test set, with a transparent imbalance-mitigation protocol (controlled oversampling + copy-paste augmentation); and (ii) StrawMind-CBAM, a YOLOv8-based detector augmented with Convolutional Block Attention Modules (CBAM) at each detection scale to improve sensitivity to the rare *Aspergillus* class. On the held-out test set, baseline YOLOv8n/s reach mAP@50 of 0.694/0.697, with near-perfect *Aspergillus* detection (mAP@50 ≈ 0.99) after imbalance mitigation, while the densely-packed *Trichoderma* class remains the hardest (≈ 0.28). Across three seeds, StrawMind-CBAM attains the best mean mAP@50 (0.6998 ± 0.0045) and the best mean *Trichoderma* detection, improving over the same-backbone YOLOv8n baseline on all three seeds at only +0.1M parameters. All data-building and training code, configs, seeds, and logs are released for full reproducibility.

Keywords: mushroom disease detection; object detection; attention; CBAM; class imbalance; YOLO; reproducible benchmark.

1 Introduction

Mushroom cultivation is highly susceptible to fungal contaminants such as *Trichoderma* (green mold) and *Aspergillus*, which spread rapidly and threaten both yield and food safety. Automated, real-time visual detection offers a low-cost monitoring path, but progress is hindered by two gaps. **First**, public mushroom-disease detection benchmarks are small, inconsistent in class definitions, and frequently lack a properly held-out test set, making reported numbers hard to compare or reproduce. **Second**, real cultivation imagery is severely class-imbalanced—in our aggregated data the rarest class (*Aspergillus*) is under-represented by roughly 29× relative to *Trichoderma* bounding boxes—yet most works neither quantify nor explicitly mitigate this.

We address both gaps with StrawMind. Our contributions are:

1. A **reproducible three-class benchmark** for mushroom fungal-disease detection with a stratified 70/15/15 split and an absolutely held-out test set, plus a documented, reviewer-friendly imbalance-mitigation protocol.

Table 1: StrawMind dataset composition (images / bounding boxes). *Aspergillus train oversampled $\times$4.

Class	Train img	Train bbox	Val img	Val bbox	Test img	Test bbox
Healthy	603	779	129	191	129	184
Trichoderma	546	3948	117	859	117	981
Aspergillus	564*	564	30	30	30	30

2. **StrawMind-CBAM**, a YOLOv8 detector with CBAM attention injected at every detection scale to boost sensitivity to small/rare lesions.

3. A **controlled empirical study** (multiple architectures, fixed seeds, identical splits) with all logs/configs released; every reported metric is read directly from training/validation logs (no estimates).

2 Related Work

Real-time object detection. One-stage detectors originate with YOLO [7]; YOLOv10 [8] removes NMS for end-to-end real-time detection, with further improvements for complex environments [10]. We use the YOLO family (v8n/s, v10n) as controlled baselines.

Attention for agricultural disease detection. Attention improves crop-disease recognition, e.g. rice-disease recognition with a novel attention mechanism [1] and passion-fruit disease detection with sparse parallel attention [2]. StrawMind-CBAM adds CBAM [9]—which extends channel attention from SE-Net [3]—at each detection scale.

Class imbalance. Focal Loss [6] and the long-tailed Equalized Focal Loss [5] are standard. As the Ultralytics framework does not expose per-class loss weighting, we adopt controlled oversampling as an equivalent, transparent mitigation.

Data augmentation. Copy-paste augmentation increases sample diversity for rare/small objects [11]; we enable it (`copy_paste` = 0.5) for the rare class.

3 The StrawMind Benchmark

3.1 Data sources and label harmonization

We aggregate two public Roboflow Universe datasets (CC BY 4.0). Labels are harmonized to three classes: `0=Healthy`, `1=Trichoderma`, `2=Aspergillus` (case-insensitive; "green mold" $\rightarrow$ *Trichoderma*).

3.2 Split and imbalance mitigation

We use a stratified 70/15/15 split (seed 42); the test set is held out absolutely and never used during training. The rare *Aspergillus* class is oversampled $\times$4 in *train only* (val/test untouched), combined with copy-paste augmentation. Table 1 summarizes the final composition.

Table 2: Main results on the held-out test set (seed 42).

Model	mAP@50	mAP@50–95	Precision	Recall
YOLOv8n	0.6937	0.5290	0.724	0.691
YOLOv8s	0.6970	0.5406	0.714	0.689
YOLOv10n	0.6873	0.5402	0.704	0.683
StrawMind-CBAM (ours)	**0.6953**	0.5385	0.712	0.684

Table 3: Held-out test results, mean±std over 3 seeds (42/123/2026). Best in **bold**.

Model	mAP@50	mAP@50–95	Trichoderma mAP@50	Params
YOLOv8n	0.6973 ± 0.0027	0.5322	0.2776 ± 0.0117	3.0M
YOLOv8s	0.6944 ± 0.0045	0.5366	0.2872 ± 0.0043	11.1M
YOLOv10n	0.6833 ± 0.0054	0.5366	0.2576 ± 0.0108	2.27M
StrawMind-CBAM (ours)	$\mathbf{0.6998 \pm 0.0045}$	0.5355	$\mathbf{0.2923 \pm 0.0058}$	3.10M

4 StrawMind-CBAM Architecture

StrawMind-CBAM builds on YOLOv8n and inserts a CBAM module after each of the three detection-scale feature maps (P3/P4/P5) immediately before the `Detect` head. CBAM applies sequential channel and spatial attention, refining features to emphasize subtle/rare lesion patterns at minimal cost. The model is warm-started from `yolov8n.pt`. StrawMind-CBAM has 3.10M parameters and 8.3 GFLOPs, only +0.1M over the YOLOv8n backbone (3.0M), confirming the attention modules add negligible cost.

5 Experimental Setup

All models are trained for 100 epochs at 640×640, batch 16, on a single NVIDIA P100 GPU, with PyTorch 2.5.1 (CUDA 12.1) and Ultralytics [4]. Seed 42 is used for the main comparison; seeds 123 and 2026 are added for models exceeding mAP@50 > 0.5 to report mean±std (3 seeds). We report precision, recall, mAP@50 and mAP@50–95 on the held-out test set. **Every metric is read directly from the Ultralytics validation logs.**

6 Results

6.1 Main comparison (seed 42, held-out test)

Table 2 reports test-set results. Baselines reach mAP@50 ≈ 0.70; imbalance mitigation yields near-perfect *Aspergillus* detection, while densely packed *Trichoderma* (≈ 7.4 boxes/image) remains hardest.

6.2 Robustness across seeds (3-seed mean±std)

Table 3 reports mean±std over three seeds (42, 123, 2026). StrawMind-CBAM attains the *highest* mean mAP@50 (0.6998) and the highest mean *Trichoderma* mAP@50 (0.2923), outperforming the same-backbone YOLOv8n baseline on *all three seeds* for overall mAP@50 ($+0.0016/+0.0004/+0.0056$; mean $+0.0025$) at a negligible cost of +0.1M parameters.

Table 4: Per-class mAP@50 (seed 42, held-out test).

Model	Healthy	Trichoderma	Aspergillus
YOLOv8n	0.810	0.276	0.995
YOLOv8s	0.806	0.293	0.992
YOLOv10n	0.797	0.271	0.994
StrawMind-CBAM (ours)	0.804	**0.287**	0.995

Table 5: Ablation on the held-out test set (seed 42). CBAM provides the dominant gain, especially on *Trichoderma*.

CBAM	copy-paste	mAP@50	mAP@50–95	Trichoderma	Aspergillus
✗	0.0	0.6937	0.5290	0.2761	0.9950
✗	0.5	0.6937	0.5290	0.2761	0.9950
✓	0.0	**0.7063**	**0.5415**	**0.3076**	0.9908
✓	0.5	0.6953	0.5385	0.2870	0.9950

We assess significance with a paired analysis over seeds. The overall mAP@50 gain of StrawMind-CBAM over YOLOv8n is consistent in sign across all three seeds (paired $t(2) = 1.61$; sign-test $p = 0.125$), indicating a *small but directionally stable* improvement rather than a strongly significant one at n=3. The clearest practical benefit is on the hardest, densely packed *Trichoderma* class (+0.0147 mean mAP@50). We report these numbers transparently and avoid over-claiming.

6.3 Per-class analysis

6.4 Ablation study

We ablate the two key components on the held-out test set (seed 42), reported in Table 5. Adding CBAM yields a clear improvement of +0.0126 mAP@50 and, most notably, +0.0315 on the hardest *Trichoderma* class (from 0.2761 to 0.3076), confirming that the attention modules—not merely augmentation—drive the gain.

7 Discussion

The results validate our imbalance-mitigation protocol: despite *Aspergillus* being the rarest class, it is detected almost perfectly, whereas the dominant but densely packed *Trichoderma* class is the principal source of error. This indicates that, in this domain, *spatial density and box scale*—not raw class frequency—drive the remaining gap, motivating the attention-based StrawMind-CBAM design. Quantitatively, StrawMind-CBAM improves over YOLOv8n on overall mAP@50 in all three seeds (mean +0.0025; paired $t(2) = 1.61$) and most strongly on *Trichoderma* (mean +0.0147); the gain is small but directionally consistent, and we deliberately refrain from over-claiming statistical significance at n=3.

8 Limitations

The original straw-mushroom images (134, with mismatched Affected/Healthy labels) are excluded from mAP computation and retained only for qualitative visualization. The aggregated source imbalance ($\approx 29\times$) limits *Aspergillus* test support (30 boxes). Results are single-GPU and on aggregated public data; field deployment requires further validation.

9 Conclusion

We introduced StrawMind, a reproducible imbalanced benchmark, and StrawMind-CBAM, a CBAM-augmented real-time detector for mushroom fungal-disease detection. With a transparent imbalance protocol, baselines already reach mAP@50 ≈ 0.70 and near-perfect rare-class detection. StrawMind-CBAM achieves the best mean mAP@50 among all tested detectors with consistent per-seed gains over its YOLOv8n backbone. All code, configs, seeds, and logs are released.

Reproducibility

Data-building notebook: `cattillallnight/strawmind-dataprep` (seed 42). Training notebooks: `strawmind-lo1` (YOLOv8n/s), `strawmind-lo2/lo2b` (YOLOv10n, StrawMind-CBAM), `strawmind-lo3` (3-seed). All metrics read from released logs.

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